AQA GCSE Physics
Topic – Magnetism & Electromagnetism
Section 1: Detailed Revision Notes

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1. Magnetic Poles & Magnetic Fields

Magnetic Poles

All magnets have two poles:

- **North pole (N)**
- **South pole (S)**

Key rule:

- Like poles repel
- Unlike poles attract

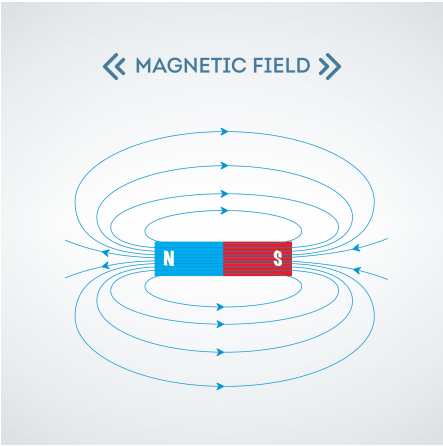
Magnetic Field

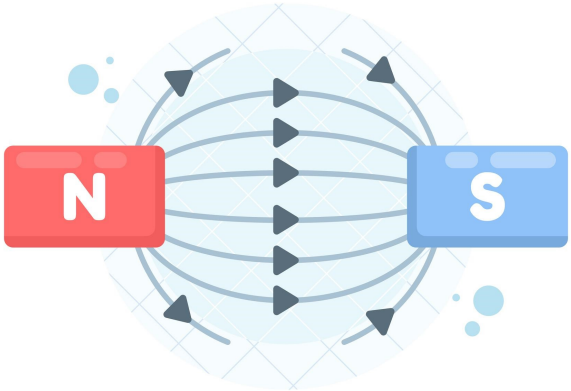
A **magnetic field** is the region around a magnet where magnetic forces act.

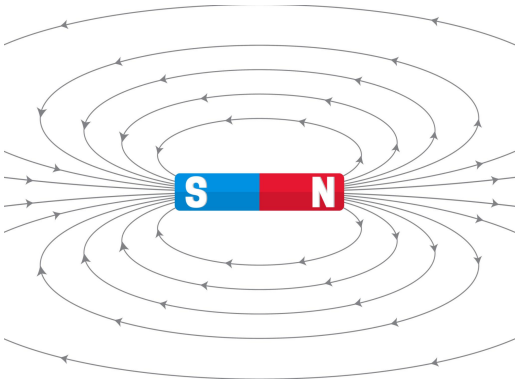
Field lines show:

- Direction from **North → South**
- Strength (closer lines = stronger field)

Bar Magnet Field Pattern







Key Points:

- Field lines never cross
- Strongest at poles
- Outside magnet: N → S

2. Electromagnets & Solenoids

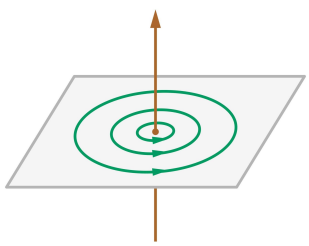
Magnetic Field Around a Current-Carrying Wire

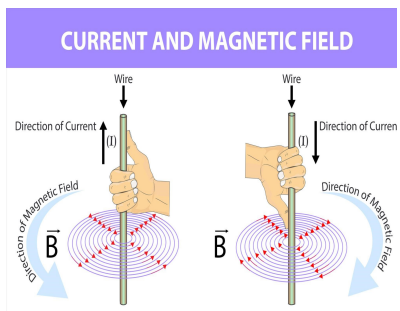
When current flows in a wire, a circular magnetic field forms.

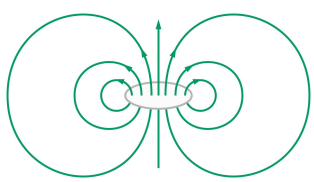
Direction is given by the **right-hand thumb rule**:

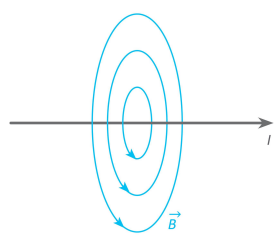
- Thumb = direction of current
- Fingers = direction of magnetic field

Magnetic Field Around a Wire







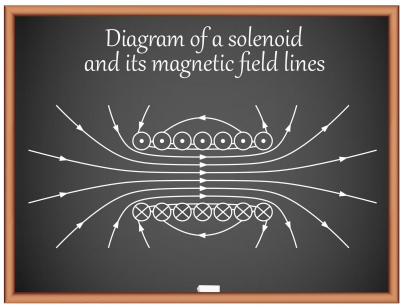


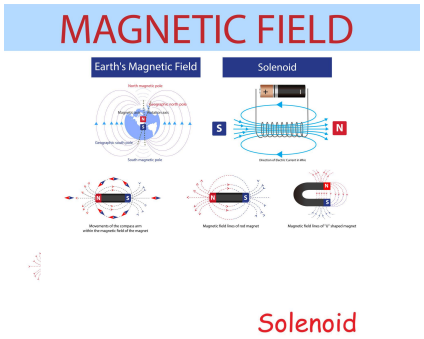
Solenoid (Electromagnet)

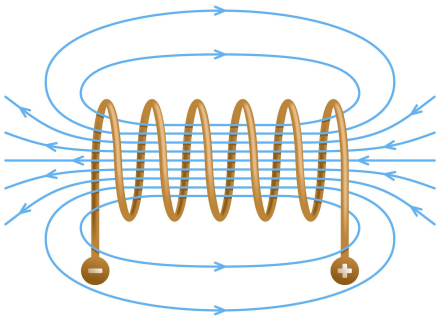
A solenoid is a coil of wire that produces a magnetic field when current flows.

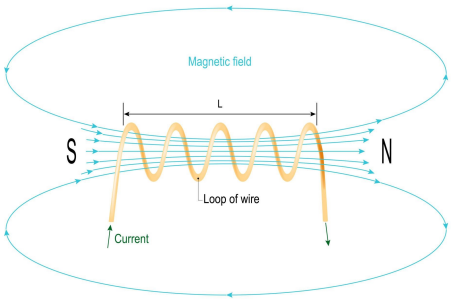
It acts like a bar magnet.

Solenoid Magnetic Field









Inside solenoid:

- Field lines are parallel
- Strong uniform field

Increasing Strength of an Electromagnet

To make it stronger:

- Increase current
- Increase number of turns
- Add an iron core

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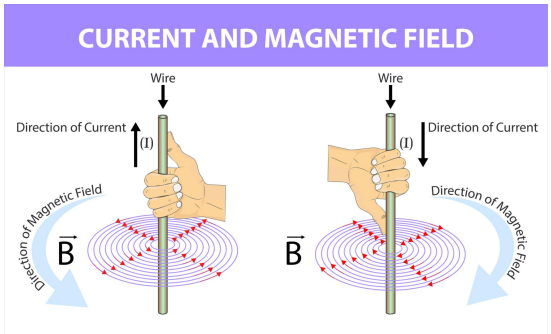
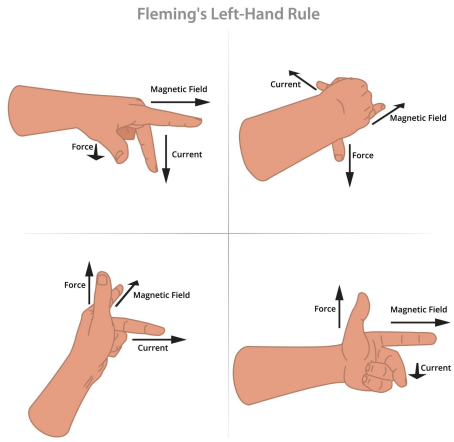
3. The Motor Effect

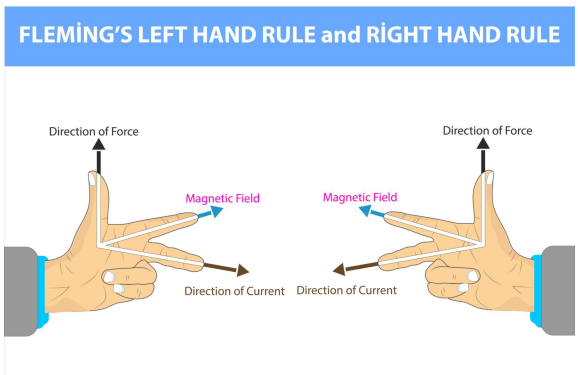
When a current-carrying wire is placed in a magnetic field, it experiences a force.

This is called the **motor effect**.

Motor Effect Diagram







Fleming's Left-Hand Rule

Used to determine direction of force.

- Thumb = Force
- First finger = Magnetic field
- Second finger = Current

Motor Effect Equation

Force = Magnetic flux density × Current × Length

$F = B \times I \times L$

Units:
F = Newtons (N)
B = Tesla (T)
I = Amps (A)
L = metres (m)

4. Electromagnetic Induction

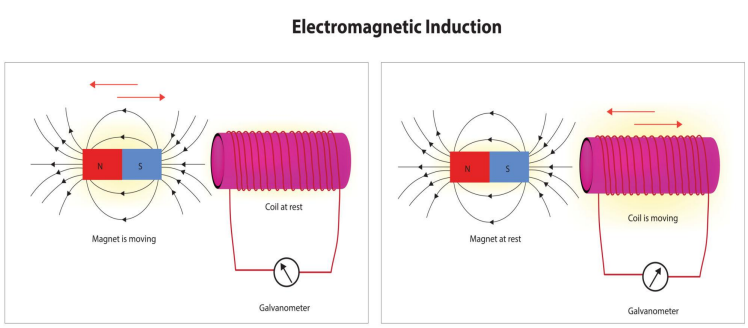
If a conductor moves in a magnetic field, a voltage is induced.

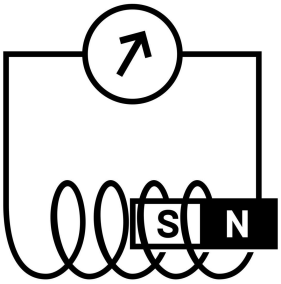
This is called **electromagnetic induction**.

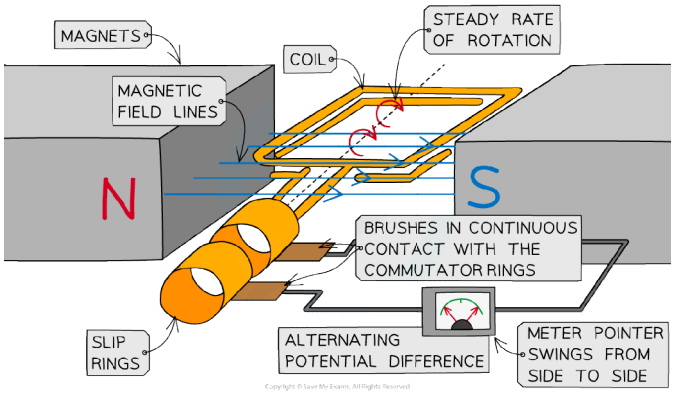
Ways to Induce a Current:

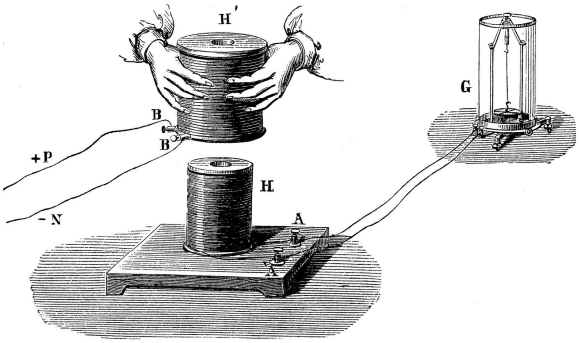
- Move wire through magnetic field
- Move magnet into coil
- Change magnetic field strength

Generator Principle Diagram









INDUCED CURRENT

Factors Affecting Induced Voltage

- Speed of movement
- Strength of magnet
- Number of turns in coil

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5. Transformers (HT Only)

Transformers change voltage using electromagnetic induction.

Work only with **AC current**.

Transformer Equation

Vp / Vs = Np / Ns

Where:
Vp = primary voltage
Vs = secondary voltage
Np = number of turns (primary)
Ns = number of turns (secondary)

Step-Up vs Step-Down

- Step-up → increases voltage
- Step-down → decreases voltage

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6. Required Practical – Investigating Magnetic Fields

Students must be able to:

- Use iron filings to show magnetic field patterns
- Use a compass to map magnetic field direction
- Draw field lines correctly

Key exam phrase:

Field lines go from **North to South** and are closer where the field is stronger.

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7. Common GCSE Exam Mistakes

- Confusing motor effect with induction
- Forgetting right-hand / left-hand rule
- Forgetting transformers only work with AC
- Mixing up North and South field direction
- Writing incorrect units (Tesla, Newton, Ampere)

Formula Summary

$F = B \times I \times L$

$V_p / V_s = N_p / N_s$

End of Section 1 – Magnetism & Electromagnetism